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324812 (24)

BE (8th Semester)

Examination, Nov.-Dec., 2013

Branch : Electrical, EEE

TILIZATION OF ELECTRICAL ENERGY

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

ote : (i) Part 'a' of each question is compulsory.

(ii) Attempt any two from the remaining part b, c or d.

1. (a) List factors that is considered for the selection of a machine for a particular application.

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(b) Draw the schematic diagram of an electric drive. Discuss the various power and contro block in it. 7

(c) Two electrical machines are considered to drive a load. The torque and speed rating are the same. The thermal limits for the machines correspond to their rated armature resitive losses. The machine 1 and 2 have the following torque to armature current relationships.

(i) $T_e \alpha i^2$

(ii) $T_e \alpha i$

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The load is twice the rated torque for half the time and zero for the other half time in load cycle. Find which machine is suitable for the application. 7

(d) Explain the reasons for using load equalization in an electrical drive. 7

Q. 2. (a) The speed controlled dc drives uses actual speed for feedback control. In the absence of tachogenerator what could be done to close the speed loop. 2

(b) State and explain the disadvantages of using a motor of wrong rating. 7

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(c) Draw the schematic diagram of a close loop speed control scheme with inner curre control loop of a drive. Enumerate the vario functions of inner current control loop ?

(d) Describe the constant torque and consta power operation of a DC/AC drive. Draw relevant diagram if any.

Q. 3. (a) Draw the speed/torque characteristic of phase I.M. with operation at :

(i) Rated voltage and supply frequenc

(ii) Supply voltage reduced by half frequency remaining constant.

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(b) Describe why the speed control of a 3ϕ induction motor using stator voltage control is considered as an inefficient method. 7

(c) A 2200 V, 2600 kW, 735 rpm, 50 Hz, 8 pole, 3 phase squirrel cage induction motor has the following parameter referred to stator :

$r_s = .075\ \Omega$, $r_2' = 0.1\ \Omega$ $x\ell s = 0.45\ \Omega$ $x'\ell r = 0.55\ \Omega$ stator winding is delta connected and consists of two section in parallel. Find the starting torque and maximum torque as a ratio of rated torque. 7

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(d) Explain why a self controlled synchronous motor is free from hunting oscillations. 7

Q. 4. (a) Define the coefficient of adhesion. 2

(b) Discuss 25 KV ac traction drive employing transformer with tap-changer and also state its advantages and disadvantages. 7

(c) Describe the advantages and drawbacks of a semiconductor converter controlled traction drives. 7

(d) Describe the various types of Electric arc welding processes. 7

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Q. 5. (a) Define the following terms : 2

(i) Candle power

(ii) Luminious intensity

(b) Discuss laws of illumination and its limitation in actual practice. 7

(c) Explain the various factors to be taken into account for designing schemes for : 7

(i) Street lighting

(ii) Flood lighting

(d) Explain how the variable speed drive allows saving of energy in pump drives. 7

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